Course Intro

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CS 0220 2022

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Overview

Who we are

- Instructor: Robert Lewis (call me Rob!)
- HTAs: Jiahua Chen, Nick Liu, Rachel Ma, Walter Zhang
- STAs: Alex Horowitz, Charles Somerville
- UTAs: 22 friendly faces :)

Our Website: Hot drinks!

- Class goals.
- Course outline.
- Meet the UTAs!
- Collaboration policy.
- Assignments, dates and deadlines.
  - Homework released Tuesdays, due following Monday at 11:59pm
  - Midterm: March 14
  - Final: May 13
- Attendance policy.
  - Lectures encouraged, not required. Recitations required.
- Recitations.
- TA hours.

Recitations

- Lectures for the "what" and the "why"; recitations for the "how"
- Required, in person(*): you'll sign up for a particular section
  - One virtual section, in case of illness/quarantine/...
- Sign up as part of HW 0

Other sites

Details, links on main site!

- EdStem: Best way to get quick answers. Key announcements there, too.
- Gradescope: Handins, homework grading.
- Overleaf (optional): LaTeX without installation.

Class goals

CS can feel like a very applied field. Why learn math?

- Problem solving
- Communication
- Collaboration

A key result of this class: you'll have a *vocabulary* for discussing certain kinds of problems that appear in many different contexts, and a toolbox of general approaches for solving them.

A vital point of computer science (academic, industry, hobbyist): *communication*.

Our expectations from you

- No mathematical background is assumed. We're not doing calculus, statistics, ...
- Approach things with an open mind.
- Try to communicate clearly and concisely.
- Respect your classmates and TAs: we're in this together.

- Let us know how you're doing!

Ethics in Discrete Math

- Two STAs this semester. Why?
- Math often seen as a “neutral” or “pure.” It’s more complicated than that.
- Math becomes relevant when it is applied to the real world. Doing so **always** requires simplifications.
- Issues arise via: (1) flawed assumptions when bridging between theory and reality, (2) ethical flaws in understanding the “end-goal” application, and more.
- Keep uses in mind. The largest employer of mathematicians in the US is the NSA, which has clear ethical implications.
- We’ll be asking you to consider potential ethical implications of the topics we cover and the importance of considering issues in advance.

Odd times odd

- Poll. How approach a problem like this one?
- Check a few cases to see if you believe it.
  $3 \times 5 = 15$, $7 \times 3 = 21$. One times anything is the same, so, if it was odd, it stays odd. So far so good.
- Go to definitions. What does odd actually mean, mathematically? A number is *odd* if it can be written $2k + 1$ for an integer k.
- Use definitions to express the problem.
  We have two odd numbers: $2k_1 + 1$, $2k_2 + 1$.
  What can we say about their product?

$$\begin{aligned} (2k_1 + 1)(2k_2 + 1) &= 4k_1k_2 + 2k_1 + 2k_2 + 1 \\ &= 2(2k_1k_2 + k_1 + k_2) + 1 \\ &= 2k_3 + 1, \end{aligned}$$

Since $k_3 = 2k_1k_2 + k_1 + k_2$ is an integer, the product is odd.

Bad "proof"

Each step must be done carefully to avoid going off the rails.

Pick any y and let $x = 2y$	x	$=$	$2y$
Multiply by $-x$	$-x^2$	$=$	$-2xy$
Add $2x^2$	x^2	$=$	$2x^2 - 2xy$
Subtract $2xy$	$x^2 - 2xy$	$=$	$2x^2 - 4xy$
Factor	$x(x - 2y)$	$=$	$2x(x - 2y)$
Cancel common terms	1	$=$	2

Conclusion: Math is over. If we can conclude $1 = 2$, we can conclude *anything*.

Proof by contradiction

- If n^2 is even, then n is even.

  We are given that n^2 is even. Let's *assume* that n is odd. Earlier in the lecture, we showed that the product of two odds is odd. That implies that n^2 must be odd. But, n^2 can't be both odd and even, so that's a *contradiction*. That means the negation of our most recent assumption *must* be true. In other words, n must have actually been even.

The basic idea is that if *any time a* happens then *b* must happen *and* we know *b* didn't happen, well, then *a* couldn't have happened either. After all, if *a had* happened, it would have made *b* happen. But, *b* didn't happen, so *a* couldn't have happened.

Deeper proof by contradiction(?)

- $\sqrt{2}$ is irrational. (In other words, it is *not* rational.)

  Let's *assume* it is rational. That means $\sqrt{2} = p/q$ where p and q are integers. Furthermore, we can assume p/q is in lowest terms so they have no factors in common. Squaring both sides, we get $2 = p^2/q^2$ or $2q^2 = p^2$. Since q^2 is an integer, and p^2 is an integer times 2, p^2 is even. By a similar argument to the one about odd squares, that means p must be even. If p is even, p^2 must be divisible by 4. Since $2q^2$ is divisible by 4, q^2 must be divisible by 2. That means both p and q are even. But, then p/q is not in lowest terms, which we had already concluded. Because we've reached a *contradiction*, we can conclude that our most recent assumption must be false. That is, $\sqrt{2}$ must *not* be rational.